

Shantanu Shastri

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EDUCATION

Carnegie Mellon University	December 2026
Master of Information Systems Management	GPA: 3.85
Manipal Institute of Technology	August 2022
Bachelor of Electronics and Communications Engineering	GPA: 3.99
<i>Relevant Coursework:</i> Data Structures, Algorithms, Distributed Systems, OOP, System Design	

SKILLS

Languages: Go (Golang), Java, Python, C++, TypeScript, JavaScript, SQL, Bash/Shell
Backend & Architecture: Financial/Payment Systems, RESTful APIs, Cloud Native, Microservices, Spring Boot
Databases & NoSQL: PostgreSQL, MySQL, Redis, Ignite, Neo4j, Milvus, Sybase, MongoDB
Infrastructure & DevOps: AWS, Azure, Docker, Kubernetes, Kafka, Zookeeper, CI/CD, TDD, JUnit

EXPERIENCE

BlackRock	January 2022 – August 2026
<i>AI Engineer Internship</i>	June 2026 – August 2026

- Financial Platforms:** Built highly reliable financial infrastructure processing 500+ post-trade exceptions/week using Go, Python, PostgreSQL, and Kafka for secure, stream-based data ingestion.
- Cloud Native Architecture:** Architected mission-critical NoSQL infrastructure linking Milvus to Neo4j via a PostgreSQL bridge, ensuring data consistency for financial systems under 100ms.
- High-Performance APIs:** Engineered a secure, scalable platform backed by rate-limited RESTful APIs handling 2K+ peak RPS, advancing Straight-Through Processing for automated financial workflows.
- Evaluation & Processing:** Created rigorous frameworks to track data drift and latency, securely processing 3.5M transactional records and handling \$14T in cash reliably and efficiently.

<i>Software Developer 2 - Associate</i>	January 2025 – July 2025
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- System Scalability:** Architected a Java Spring Boot financial data framework and TTL Azure microservices, automating 1.5M+ client touchpoints and reducing infrastructure costs by 40%.
- Latency & NoSQL:** Engineered a distributed read-through NoSQL cache with LRU eviction to address high-latency APIs, scaling concurrent requests and cutting response times from 450ms to 150ms.

<i>Software Developer 1 - Analyst</i>	July 2022 – December 2024
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- Test-Driven Development (TDD):** Championed TDD by building Azure DevOps CI/CD pipelines with comprehensive JUnit tests, ensuring thoroughly tested code and reducing deployment time by 40%.
- Database Optimization:** Took end-to-end ownership of database optimization, refactoring PostgreSQL schemas on tables exceeding 50M+ rows to achieve a 37% reduction in query latency.
- Distributed Systems:** Resolved singleton bottlenecks by engineering a sharded server environment with Zookeeper, boosting system throughput to 10K+ RPS for mission-critical services.

PROJECTS

High-Performance Modular Booking Architecture

- Engineered a decoupled microservices architecture using Apache Kafka, seamlessly handling bursts of 100,000+ real-time requests with Cloud Native containerized Docker deployments.

Zulip Open-Source Engine Extensions

- Built a highly reliable, high-throughput message engine using Node.js and Redis to securely process live data streams, ensuring continuous availability and zero data loss.